Yuning Wu

Machine Learning, Robotics, Human-Robot Interaction, Computational Design
yuningw@andrew.cmu.edu • 412-626-2495 • https://ice-5.github.io/

I enjoy using AI methods to enable a humane, intelligent robotic future.

EDUCATION /

Carnegie Mellon University. Pittsburgh, USA. 2019.8-2024.5 (Expected)
PhD Candidate, Computational Design
Co-Advisors: Dr. Daniel Cardoso Llach, Dr. Jean Oh

Carnegie Mellon University. Pittsburgh, USA. 2019.8 - 2021.5
MS, Machine Learning
Research Advisor: Dr. Katerina Fragkiadaki

Tsinghua University. Beijing, China. 2010.8 - 2017.5
BArch MArch, Architecture
Focus Area: Generative 3D Design, Robotic Prototyping, Digital Fabrication
The experience equips me with knowledge and courage to build challenging things

INDUSTRY EXPERIENCE /

Research Collaborator 2021.8 - 2022.5
Autodesk Research, AI Lab & Robotics Lab. Remote, USA.
Collaborators: Dr. Jieliang Luo (Principal AI Scientist), Dr. Hui Li (Sr. Principal Scientist)
- Work on Inverse Reinforcement Learning (IRL) in contact-rich robotic manipulation.
- Research published and presented on **ICRA 22**. [P2]

Research Intern 2021.5 - 2021.8
Autodesk Research, AI Lab & Robotics Lab. Remote, USA.
Supervisors: Dr. Jieliang Luo, Dr. Hui Li

CTO, Co-founder 2017.12 - 2019.8
Metascope Technology. Beijing, China.
- Build a vision-based style recommendation system for commercial architects.
- Use segmentation methods to preprocess large building image datasets.
- Customize DNN models for recognizing visual architectural styles.
- Use statistical machine learning methods to draft efficient offline profiling models.

RESEARCH EXPERIENCE /

Technical Lead 2021.8 - 2023.5
Rethinking Automation in Construction: Towards Reinforcement Learning Driven Robotically-Supported Collaborative Construction (RSCC) [Preview]
Advised by Dr. Daniel Cardoso Llach, Dr. Jean Oh
- Envision how AI and robotics can support existing labor-intensive workflows.
- Build a companion rover can deliver tools and materials to workers in a complex, messy, dynamic construction environment.
- Combine RL-driven social navigation methods with sampling-based methods to form a safe and efficient navigation strategy among human workers.
- Design the complete ROS-based, multi-sensor robotic system from a simple chassis.
- Deploy SoTA LiDAR Odometry and Mapping (LOAM) methods for 3D site perception.
- Combine vision-based (YOLOv7) and LiDAR-based (PointPillars) methods to robustly detect and track human workers' moving trajectories.
- Use NVIDIA Omniverse Isaac to build a customized simulation environment for RL.
- Optimize real-time edge deployment of DNN models with GPU acceleration.
- Research accepted to **CSCW 23**. [P1]

Independent Research 2021.8 - WIP
Teaching AI Spatial Sense by Playing with LEGOs [Link]
Advised by Dr. Katerina Fragkiadaki
- Build a GNN-based physics engine to infer the stability of a stacked LEGO structure.
- Use hierarchical reinforcement learning (HRL) to search for valid building patterns.
- Partial demonstrations are available.

Algorithm Lead — 2020.5 - 2020.8

Semi-Autonomous Construction in Dynamic Environments: A Reinforcement Learning Approach Towards Scalable and Robust Multi-Agent Drone Construction [Link] [Demo]

Advised by Dr. Daniel Cardoso Llach

- Raise a semi-autonomous framework for building bespoke, real-world structures with a fleet of drones.
- Train and deploy a decentralized, RL-based multi-agent collision avoidance algorithm.
- Customize a Sim2Real pipeline using Unity MLAgents.
- Research published and presented on **ACADIA 20**. [P3]

Member of the System Engineering Team — 2020.8 - 2020.12

MoonRanger [Link]

Advised by Dr. William (Red) Whittaker, Dr. Heather Jones

- Participate in the system engineering of an autonomous, suit-case-sized exploration rover searching for water and ice on lunar surface.
- The launch is scheduled for 2023 through the NASA CLPS program.

PUBLICATION /

Y. Wu, E. Erdolu, J. Wei, D. Cardoso Llach. **"Robot in the Loop:" From Construction Automation to RL-Driven Robotically-Supported Cooperative Construction**. ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW 2023). Accepted, pending publication. [P1]

Y. Wu, J. Luo, H. Li. **Learning Dense Rewards with Temporal Variant Self-Supervision**. International Conference on Robotics and Automation (ICRA 2022), Reinforcement Learning for Contact-Rich Manipulation Workshop. [Link] [P2]

Z. Fang, **Y. Wu**, A. Hassonjee, A. Bidgoli, D. Llach. **Towards an Architectural Framework for Distributed, Robotically-Assisted Construction: Using Reinforcement Learning to Support Scalable Multi-Drone Construction in Dynamic Environments**. Conference of the Association for Computer-Aided Design in Architecture (ACADIA 2020). [Link] [P3]

O. Akin*, **Y. Wu***. **State Space Paradox of Computational Research in Creativity**. Conference on Computational and Cognitive Models of Creative Design (HI 2019). [P4]

HONOR & AWARD /

Presidential Fellowship. Carnegie Mellon University, Tata Consultancy Services — 2022
Manufacturing Futures Institute Fellowship. Carnegie Mellon University — 2022
Autodesk Research Merit Scholarship. Autodesk Research, Carnegie Mellon University — 2021
Manufacturing PA (MPA) Fellowship. State of Pennsylvania — 2021
GSA-Provost GuSH Grant. Carnegie Mellon University — 2020
Lego Scholarship. Lego Foundation — 2019
Distinctive Graduation Thesis. Tsinghua University — 2017
Siyuan Exceptional Student Fellowship. Tsinghua University — 2010-2014

INVITED TALK /

Tata Innovation Forum 2023. TCS, Cornell Tech (Pending) — 2023.6
Autodesk AI Lab Research Seminar. Autodesk Research (Pending) — 2023.6
CMU MFI Research Seminar Series. CMU MFI — 2022.11
Situating Research. CMU 48-620 Graduate Seminars — 2022.10
Inquiries into Machine Learning and Design. CMU 48-770 Graduate Seminars — 2021.11

SKILL /

Machine Learning. PyTorch, Tensorflow+Keras, Spark, Numpy, Scipy, Matplotlib
Robotics. ROS, NVIDIA Omniverse Isaac, Unity+MLAgents, Mujoco, OpenAI Gym
Programming. Python, C++, C, C#
3D Modeling and Prototyping. AutoCAD, Rhino+Grasshopper, Revit+Dynamo

RELEVANT COURSE /

10701-Introduction to Machine Learning (Ph.D.) • 10716-Advanced Machine Learning • 16720-Computer Vision • 10605-Machine Learning with Large Dataset • 10725-Convex Optimization • 10708-Probabilistic Graphical Models • 10703-Deep Reinforcement Learning and Control • 16865-Space Robotics • 36700-Probability and Mathematical Statistics